

FA 2

Exercise 3.49

$$\begin{aligned} 3.49 \\ \sum_{j=1}^N (x_j - 1)^2 &= \sum_{j=1}^N x_j^2 - 2 \sum_{j=1}^N x_j + N \\ &= \sum_{j=1}^N (x_j^2 - 2x_j + 1) \\ &= \sum_{j=1}^N (x_j^2 - 2x_j + 1) \\ &= \sum_{j=1}^N x_j^2 - 2 \sum_{j=1}^N x_j + \sum_{j=1}^N 1 \\ &= \sum_{j=1}^N 1 = N \\ &= \sum_{j=1}^N (x_j - 1)^2 = \sum_{j=1}^N x_j^2 - 2 \sum_{j=1}^N x_j + N \end{aligned}$$

Exercise 3.51

$U = c(3, -2, 5)$

$V = c(-4, -1, 6)$

$a = \text{sum}(U * V)$

$b = \text{sum}((U + 3) * (V - 4))$

$c = \text{sum}(V^2)$

$d = \text{sum}(U) * (\text{sum}(V))^2$

$e = \text{sum}(U * V^2)$

$f = \text{sum}(U^2 - 2 * V^2 + 2)$

$g = \text{sum}(U / V)$

```
results = data.frame(  
  Part = c("a", "b", "c", "d", "e", "f", "g"),  
  Value = c(a, b, c, d, e, f, g)  
)  
print(results)
```

Output

	Part	Value
1	a	20.000000
2	b	-37.000000
3	c	53.000000
4	d	6.000000
5	e	226.000000
6	f	-62.000000
7	g	2.083333

Exercise 3.90

```
a = c(3, 5, 8, 3, 7, 2)
b = c(28.5, 73.6, 47.2, 31.5, 64.8)

geometricMeanA = prod(a)^(1/length(a))
geometricMeanB = prod(b)^(1/length(b))

geometricMeanA
geometricMeanB
```

Output

```
[1] 4.140681
[1] 45.8258
```